

BHOOMIKA MONTHY RAJASHEKAR

☎ (857) 919-9906 ✉ monthyrajashekar.b@northeastern.edu  [linkedin.com/in/bhoomika-rajshekar](https://www.linkedin.com/in/bhoomika-rajshekar)
 github.com/bhoomi02-ai  [portfolio](#)

Education

Northeastern University

Master of Science in Artificial Intelligence

Sep 2026 – Dec 2027

Boston, MA

Boston University

Master of Science in Artificial Intelligence (transferred)

Sep 2025 – July 2026

Boston, MA

GSSSIETW – First Class Distinction

Bachelor of Engineering in Computer Science, AI & ML | GPA: 3.5

Jan 2021 – May 2025

India

Technical Skills

LLMs and Generative AI: GPT-4, Claude, Llama, Transformer Architectures, Fine-tuning, Prompt Engineering, RAG, LangChain, Hugging Face Transformers

Machine Learning: Supervised/Unsupervised Learning, Deep Learning, Neural Networks, Computer Vision, NLP, Model Evaluation, Hyperparameter Tuning, Feature Engineering

ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Keras, XGBoost, Pandas, NumPy, OpenCV

MLOps and Deployment: Docker, Kubernetes, Git, CI/CD, FastAPI, Flask, MLflow, Model Monitoring, REST APIs

Cloud and Data: AWS (SageMaker, Lambda, S3, EMR), Azure, GCP, MySQL, PostgreSQL, MongoDB, Snowflake, Vector Databases, ETL Pipelines, Apache Spark

Programming Languages: Python, SQL, R, Java, Unix/Linux Scripting

Data Visualization: Matplotlib, Seaborn, Tableau, Streamlit

Experience

Boston University

Jan 2026 – May 2026

Teaching Assistant – Theory and Applications of Large Language Models

Boston, MA

- Designed and delivered 15 weekly lab sessions on transformer architectures, attention mechanisms, fine-tuning, and RAG for 40+ graduate students, resulting in measurably higher project quality scores across NLP and LLM coursework
- Evaluated 100+ machine learning assignments and mentored 10+ student teams building production-ready LLM applications using LangChain and Hugging Face, reinforcing responsible AI and bias mitigation practices

Eli Lilly and Company

Feb 2025 – Aug 2025

Cloud Data Engineer

Bangalore, India

- Reduced manual clinical data processing time by 40% by architecting a secure patient data management platform for 50,000+ records using React.js, GraphQL, AWS Lambda, and S3, streamlining physician and scientist workflows by 30%
- Cut laboratory reporting time from 2 hours to 15 minutes (88% reduction) by engineering real-time monitoring dashboards in R and Shiny that processed 500GB+ of operational data monthly across 15+ global sites, improving overall efficiency by 35%

Cognizant

Jun 2024 – Jul 2024

Machine Learning Intern – GenAI Track

Bangalore, India

- Decreased average customer support response time by 60% (from 10 to 4 minutes) by designing and deploying a real-time NLP-powered AI chatbot handling 1,000+ daily queries using Microsoft Azure, OpenAI APIs, Python, and Dialogflow, increasing user satisfaction scores by 45%

Projects

Cross-Lingual Hallucination Drift in LLMs | [GitHub](#)

Boston University, 2026

NLP | LLM Evaluation | Statistical Analysis | Python | Hugging Face | Streamlit

- Reduced hallucination rate by 92% in low-resource languages by designing a six-cell multilingual evaluation framework for Aya Expanse 8B across English, Spanish, Italian, and Swahili on TruthfulQA and XCOPA benchmarks, revealing strong task-dependent drift ($\chi^2 = 45.44$, $p < 0.001$)
- Isolated task type as primary driver of hallucination drift via a within-Italian confound-controlled experiment; validated using dual-judge pipeline (GPT-4o-mini, Claude, human annotators, $\kappa = 0.662$) and deployed interactive Streamlit dashboard, paper submitted for publication

Ghost-Free SLAM: Static Scene Recovery in Dynamic Environments | [GitHub](#)

Boston University, 2026

Computer Vision | Deep Learning | Python | YOLOv8 | ORB-SLAM3 | OpenCV

- Improved localization accuracy by 92.2% in high-dynamic and 79.7% in moderate-dynamic scenes by developing a modular instance segmentation pipeline using YOLOv8 to filter dynamic human regions before ORB-SLAM3 feature extraction, evaluated on TUM RGB-D benchmark
- Recovered occluded static scene geometry without modifying the SLAM backend by implementing a depth-consistent multi-frame geometric reprojection module with 6-window temporal aggregation, achieving 14% additional ATE improvement in moderately dynamic sequences

Leadership, Publications & Achievements

Research Publication: Authored and submitted paper on cross-lingual hallucination drift in LLMs; conducted multilingual NLP experiments across 4 languages and 900 evaluation instances [Submitted for Publication]

GRS Campus Ambassador, Boston University: Represented Graduate School of Arts & Sciences; organized events and mentored international graduate students

Campus Lead, Google Developer Groups: Led 100+ member developer community; organized 15+ AI/ML workshops and mentored students in machine learning

NGO Volunteer: Taught English to 50+ underprivileged students; organized fundraising campaigns generating over 100,000 INR